

Three-Dimensional Fiber Network Structure-Reinforced Composite Solid-like Electrolyte for Lithium-Metal Batteries

Wanqing Fan, Ying Huang,* Meirong Gu, Xu Lu, Kaihang She, and Zheng Zhang*



Cite This: *Energy Fuels* 2023, 37, 16143–16149



Read Online

ACCESS |



Metrics & More

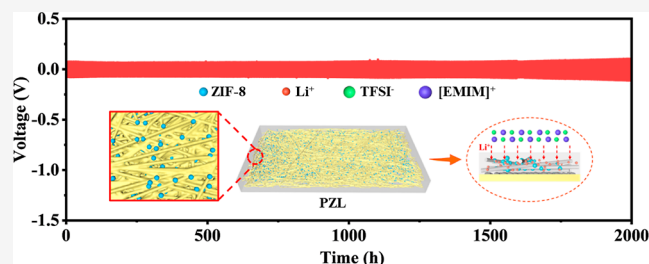


Article Recommendations



Supporting Information

ABSTRACT: Solid-like electrolytes (SLEs) based on metal–organic frameworks (MOFs) have attracted much attention due to their combination of the advantages of solid electrolytes and liquid electrolytes (LEs). Most MOF-based electrolytes using organic LEs cannot fundamentally solve the safety problems of lithium-metal batteries (LMBs), and the ionic conductivity (IC) and mechanical properties of the electrolytes should be improved. In this paper, Li-containing ionic liquids-impregnated polyvinylidene fluoride-hexafluoropropylene fiber network-loaded ZIF-8 nanoparticle electrolyte film was designed. The SLEs have excellent properties such as an IC of $2.94 \times 10^{-3} \text{ S cm}^{-1}$ at 30°C , electrochemical stability window (5.37 V), Li^+ transference number (0.56), and mechanical strength. In addition, the LFP/Li battery assembled with this electrolyte has a capacity retention rate of 93.03% after 200 cycles at 0.2C. The assembled Li symmetrical battery exhibits excellent Li plating/stripping performance for 2000 h at $0.2 \text{ mA cm}^{-2}/0.1 \text{ mA h cm}^{-2}$.



1. INTRODUCTION

Lithium-metal batteries (LMBs) are widely considered the next generation of energy storage devices due to their ultrahigh theoretical capacity (3860 mA h g^{-1}), low redox potential (-3.04 V vs the standard hydrogen electrode), and small density (0.59 g cm^{-3}).^{1–4} However, the nature of Li metal is active, and it is easy to grow Li dendrites in combination with traditional liquid electrolytes (LEs), resulting in instability of the electrolyte/Li interface and causing serious safety problems such as leakage and fire.^{5–7}

Metal–organic frameworks (MOFs) have become a new hot topic in electrolytes due to their excellent properties, such as their high specific surface area and adjustable pore structure.^{8–13} Among them, inorganic zinc ions (Zn^{2+}) and 2-methylimidazole (2-MeIm) are combined by covalent coordination bonds to form a crystalline microporous molecular sieve imidazolate framework (ZIF-8) with high porosity and excellent chemical and thermal stability, which makes it play a huge potential in many fields.^{13–17} Recently, many studies have focused on the addition of Li-ion conductors and inorganic fillers to the polymer matrix and the obtained composite solid electrolytes.^{18–20} However, the lower ionic conductivity (IC) of the composite solid electrolyte limits the LMB operation at room temperature. The MOF membrane-based solid-like electrolytes (SLEs) exhibited improved IC and the Li^+ transference number contributed to the organic LE adsorption, but their flammability causes potential safety risks.^{21,22} Therefore, it is necessary to develop an optimized electrolyte material that takes advantage of both SLEs and LEs to achieve excellent

electrochemical performance.^{23–25} This will show promising applications and facilitate the commercialization of LMBs.

Herein, polyvinylidene fluoride-hexafluoropropylene (PVDF-HFP) and ZIF-8 composite fiber membranes were prepared by the electrospinning process so that ZIF-8 can be uniformly dispersed in the entire PVDF-HFP fiber, which is different from the previously reported modified separators of MOFs that only exist on one or both sides of the surface.^{26–31} The porous ZIF-8 has a three-dimensional open framework that provides many direct contact points for the active material at the atomic level when impregnated with lithium-containing ionic liquids (Li-ILs), facilitating Li^+ transport.^{24,32,33} The resulting PZL electrolyte exhibits excellent IC ($2.94 \times 10^{-3} \text{ S cm}^{-1}$, 30°C), a high electrochemical stability window (5.37 V), and a Li^+ transference number (0.56). In addition, the distribution of ZIF-8 in PZL helps to regulate the uniform deposition of the Li metal. At $0.2 \text{ mA cm}^{-2}/0.1 \text{ mA h cm}^{-2}$, the Li/PZL/Li battery exhibits highly stable Li plating/stripping within 2000 h. Based on the good compatibility between PZL and Li metal, the assembled $\text{LiFePO}_4/\text{Li}$ battery has good rate performance and cycle performance.

Received: August 8, 2023

Revised: September 25, 2023

Published: October 10, 2023



2. RESULTS AND DISCUSSION

The preparation process of the PZL electrolyte is schematically shown in Figure 1. First, ZIF-8 nanoparticles were synthesized

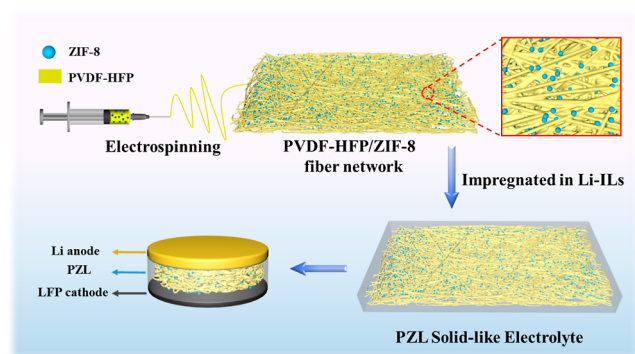


Figure 1. Schematic diagram of PZL solid electrolyte preparation.

by solution static methods, and they exhibited a rhombic dodecahedral structure (Figure S1) with a high specific surface area ($1343 \text{ m}^2 \text{ g}^{-1}$) and a pore size concentrated at 1.72 nm (Figure S2). Second, PVDF-HFP and ZIF-8 were dispersed in a mixed solution of acetone and DMF to obtain a spinning solution, and PVDF-HFP/ZIF-8 fibrous membranes were prepared by an electrostatic spinning process. Finally, the obtained fibrous membrane was impregnated in Li-ILs to obtain the PZL SLEs. ZIF-8 particles were uniformly distributed on the PVDF-HFP fibrous membrane, which made the composite material more affinity to Li-ILs and contributed to the rapid transport of Li^+ . In addition, the microporous size of ZIF-8 can restrict the migration of EMIM^+ and TFSI^- , promoting the migration of Li^+ .^{10,24,34} Meanwhile, based on the Lewis acid–base theory, TFSI^- is a Lewis base,

while Zn^{2+} in ZIF-8 behaves as a Lewis acid, providing ZIF-8 with the feature of inhibiting the movement of TFSI^- in the PZL electrolyte and releasing more Li^+ to participate in the ion transport.³⁵

The morphologies of the prepared samples were observed. Figure 2a shows that PVDF-HFP forms a good fiber structure. While the obtained PVDF-HFP/ZIF-8 fiber structure was dispersed with ZIF-8 particles (Figure 2b,c). The distribution of ZIF-8 facilitates the uptake of Li-ILs, which improves the IC of the electrolyte. Observation of the cross-sectional and surface field emission scanning electron microscopy (FESEM) images of the electrolyte revealed (Figure 2d,e) that the obtained PZL had a thickness of $62.5 \mu\text{m}$, and the impregnated Li-ILs completely encapsulated the fibers. In addition, energy-dispersive X-ray spectroscopy (EDS) mapping (Figure 2f) of the PZL cross-section showed that the elements C, O, N, F, S, and Zn were uniformly distributed in the electrolyte, confirming that the ZIF-8 particles were embedded inside the PVDF-HFP fibers. Furthermore, Figure 2g shows that the PZL SLEs have good flexibility and can be bent and folded. X-ray diffraction (XRD) tests were also performed to investigate the crystalline behavior of the PVDF-HFP/ZIF-8 fiber membranes. As shown in Figure 2h, all the diffraction peaks of PZL corresponded well to the characteristic peaks of ZIF-8 compared with those of pure PVDF-HFP, which further confirmed the successful composite of ZIF-8 particles with a fiber network.

Differential scanning calorimetry (DSC) was used to characterize the phase transition behavior of the PL and PZL electrolyte films. As shown in Figure 3a, the melting temperature (T_m) of PL appeared at 65.4°C , while the T_m of PZL loaded with ZIF-8 decreased to 60.9°C . The thermal stability of the SLEs was examined using a thermogravimetric analyzer. As shown in Figure 3b, there was residual solvent loss

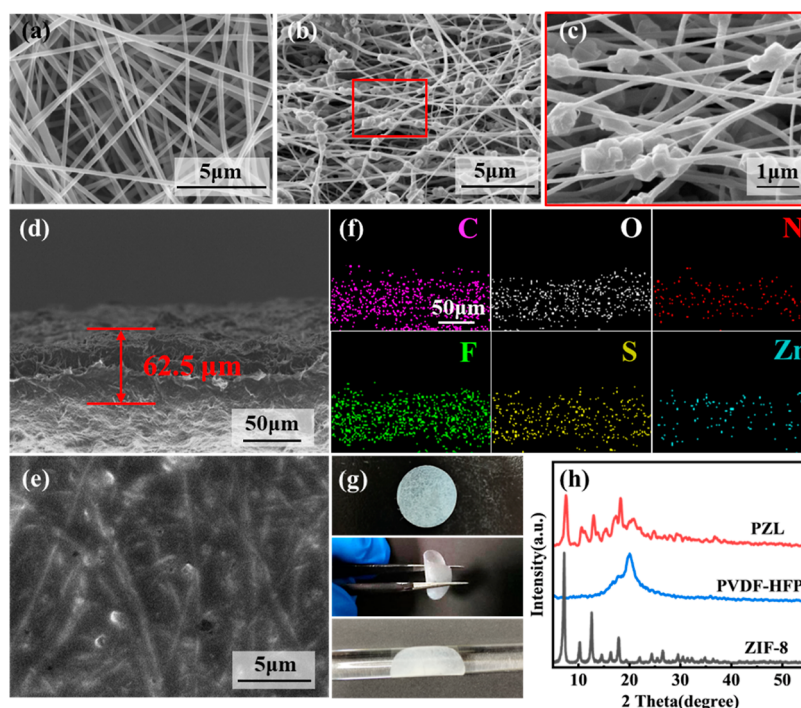


Figure 2. FESEM image of (a) PVDF-HFP, (b,c) PVDF-HFP/ZIF-8, (d) cross-section, and (e) surface morphology of PZL. (f) EDS mapping of the PZL cross-section. (g) Optical photographs of PZL. (h) XRD spectra.

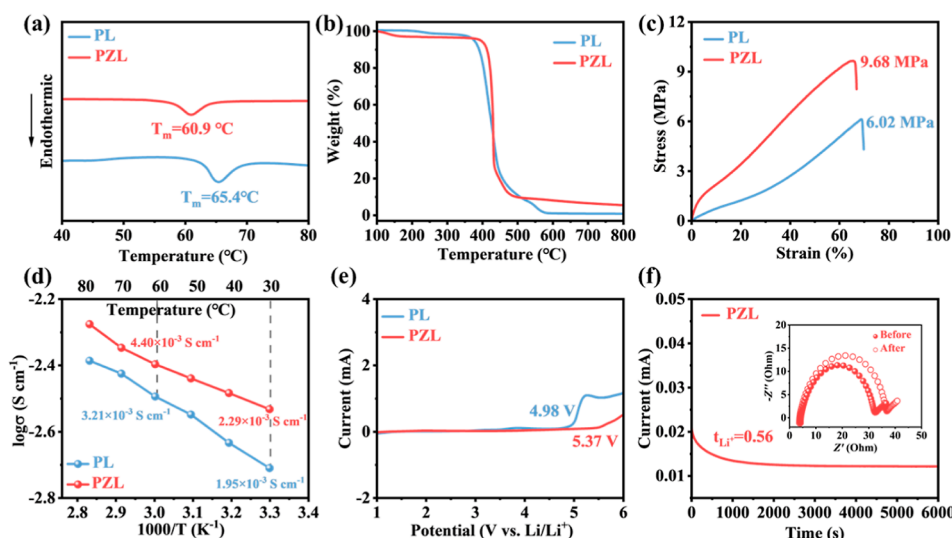


Figure 3. (a) DSC curves. (b) TG curves. (c) Stress–strain curves. (d) IC. (e) Electrochemical stability window. (f) Li^+ transference number of PZL.

until 350 °C. The PL electrolyte started to decompose at 358 °C, while the PZL started to decompose at around 400 °C, indicating that ZIF-8 improved the thermal stability of the PVDF-HFP fiber matrix, which is beneficial for high-safety LMBs. Moreover, the stress–strain curves (Figure 3c) show that ZIF-8 significantly enhances the mechanical strength of PVDF-HFP polymer, making the tensile strength of PZL as high as 9.68 MPa. This is beneficial for the electrolyte to resist the growth of Li dendrites and enhance the interface stability.^{13,36}

IC is a crucial factor in SLE applications. Figure 3d demonstrates the IC values of PL and PZL from 30 to 80 °C, respectively. Overall, the ICs of both PZLs are higher than those of PL. Figure S3 shows the Nyquist plots of symmetric cells from 30 to 80 °C. The IC of PZL is as high as $2.29 \times 10^{-3} \text{ S cm}^{-1}$ at 30 °C, which is much higher than PL ($1.95 \times 10^{-3} \text{ S cm}^{-1}$). Figure S4 shows that the Nyquist points of the PZL measured at room temperature from the first day only to the seventh day overlapped well. LSV curves (Figure 3e) show that PZL has a wide electrochemical stabilization window (5.37 V). In addition, the Li/Li symmetric cells were assembled to examine the Li^+ transference number (t_{Li^+}) of the electrolyte. As shown in Figure 3f, the t_{Li^+} of the PZL was 0.56, which was much higher than that of PL (0.27, Figure S5). This is attributed to the limiting effect of the distribution of ZIF-8 in the fibers on $[\text{EMIM}]^+$ and $[\text{TFSI}]^-$, with the smaller-sized Li^+ being less affected.^{37,38} In addition, the interconnected structure of the fiber network also provides a fast transfer channel for Li^+ transport,^{39–42} resulting in excellent electrochemical properties as described above.

Li/Li symmetric cells were assembled to monitor the plating/stripping process of Li metal for evaluating the interfacial stability of SLEs/Li. As shown in Figure 4a, the Li/PZL/Li battery exhibited a smooth and stable profile for 2000 h. Li/PZL/Li exhibits larger interfacial impedance (Figure S6) and overpotential (Figure 4b) due to the lower IC and Li^+ transference numbers of the PL electrolyte.^{24,27} In addition, the improved mechanical strength of PZL helps to inhibit Li dendrite growth. In contrast, the Li/PL/Li cells showed unstable voltage during the plating/stripping process and the polarization gradually escalated after 700 h until short-

circuiting after 1725 h. This indicates the instability of the PZL/Li interface and the growth of Li dendrites penetrating the electrolyte and causing a short circuit. Next, the Li surface morphology after cycling was compared by FESEM. Li/PL/Li cells have obvious Li dendrite growth and a rough Li surface after 300 h of cycling (Figure 4c). After cycling to 600 h, Li dendrites grow further and the surface roughness of Li metal increases. This will cause the PL/Li interface to be extremely unstable and the overpotential will increase. On the contrary, as shown in Figure 4d, the Li/PZL/Li cell has a smooth and flat Li surface after 600 h of cycling, which further confirms that the PVDF-HFP/ZIF-8 fibers guided the uniform deposition of Li^+ , resulting in uniform and stable Li plating/stripping at the interface. Furthermore, the interfacial phases of the PZL and Li anodes were analyzed and studied by X-ray photoelectron (XPS) spectroscopy. As shown in Figure 4e, an interfacial phase consisting of organic phase ($\text{C}-\text{OCO}_2\text{Li}$) and inorganic phases (Li_3N , LiF , Li_2SO_3 , and Li_2O) was detected after cycling the Li/PZL/Li symmetric cell for 300 h. The multicomponent interfacial phase is reported to be the key to stabilizing the compatibility of the electrode with the electrolyte. Among them, LiF is an important component, which has a high IC and plays an important role in improving the stability of the electrode–electrolyte interface.^{43–45}

Figure 5 depicts the plating/stripping process for Li/Li symmetric cells. For Li/PL/Li cells, the PVDF-HFP fibers are able to have sufficient fiber gaps to store Li-ILs. However, during the plating/stripping process, the anions were brought into the transport process, which created an opposite force on the migration of Li^+ . Meanwhile, the growth of Li dendrites causes the instability of the PL/Li interface, resulting in poor electrochemical performance of PL. On the contrary, in Li/PZL/Li cells, first, ZIF-8 promotes Li^+ transport because it can immobilize anions.⁴⁶ Second, ZIF-8 enhances the mechanical strength of SLEs, which can effectively hinder the growth of Li dendrites, and the PZL electrolyte with a high IC and Li^+ transference number can guide the Li to be uniformly deposited, effectively improving the SLEs/Li interface.^{42,47,48} Therefore, ZIF-8 particles are believed to positively improve the plating/stripping behavior of Li metal.

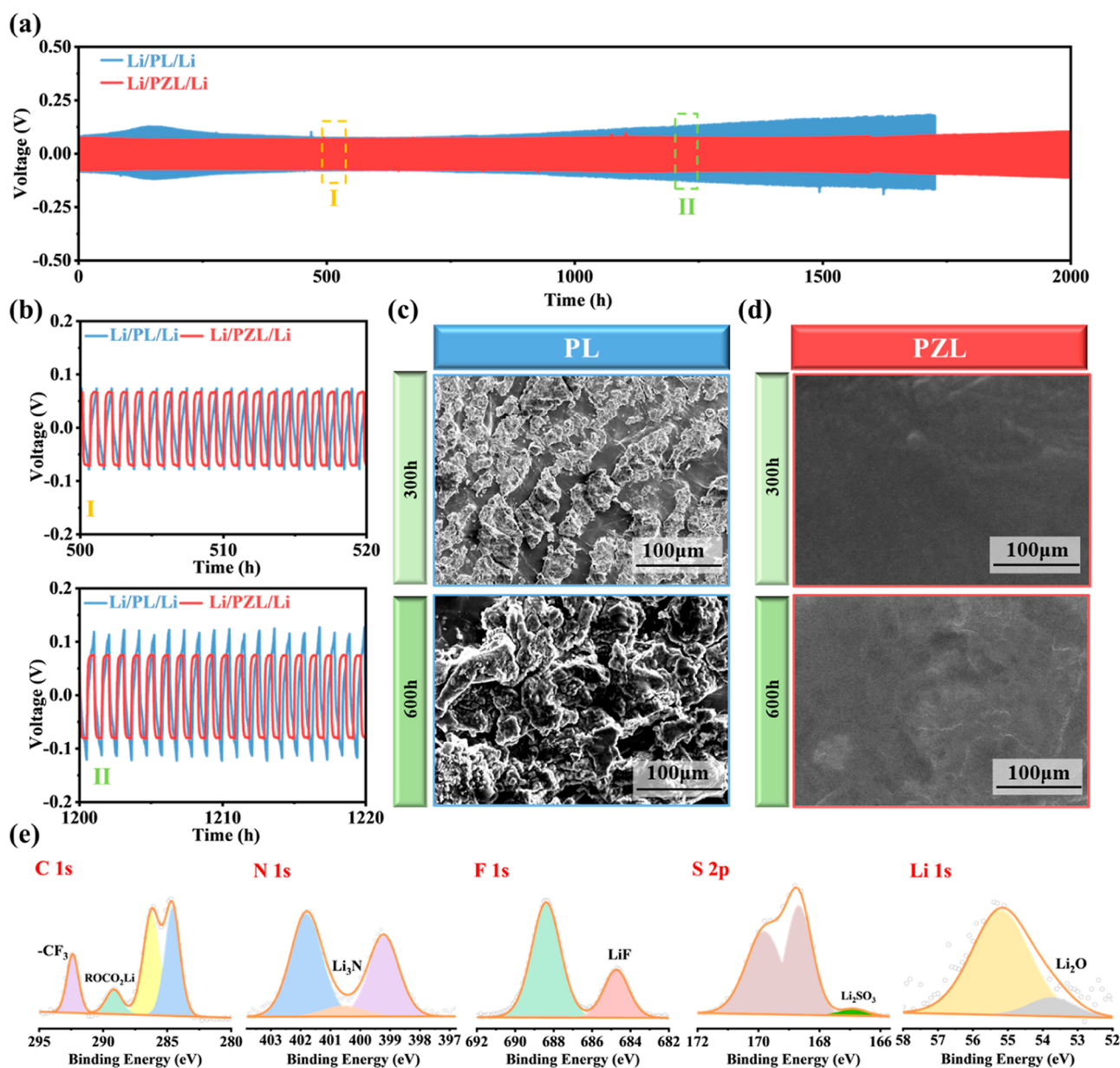


Figure 4. Electrochemical performance of Li/Li symmetric batteries. (a) Voltage profiles of Li plating/stripping for Li/PL/Li and Li/PZL/Li symmetrical cells at a current density of 0.2 mA cm^{-2} with a capacity of 0.1 mA h cm^{-2} and (b) I and II are partial enlargements. Li surface morphology before and after cycling of (c) Li/PL/Li and (d) Li/PZL/Li cells. (e) XPS spectra of C 1s, N 1s, F 1s, S 2p, and Li 1s of PZL electrolyte after 300 h of cycling.

LFP/SLEs/Li cells were assembled to further demonstrate the improved electrochemical performance of PZL SLEs. As shown in Figure 6a, the discharge capacity of the LFP/PL/Li cell was higher than that of the LFP/PZL/Li cell in the range of 0.2 to 1.0C. The discharge specific capacities of the LFP/PZL/Li cell were 153.2, 133.3, and $124.5 \text{ mA h g}^{-1}$ at rates of 0.2, 0.5, and 1.0C. When the current density was brought back to 0.2C, the discharge specific capacity of the LFP/PZL/Li cell could reach $152.2 \text{ mA h g}^{-1}$, showing excellent rate performance. However, the average discharge specific capacities of the LFP/PZL/Li cells were 106.3, 78.04, and 23.8 mA h g^{-1} at rates of 0.2, 0.5, and 1.0C. Meanwhile, according to the charge/discharge voltage distribution curves shown in Figure 6b, it can be seen that at 0.2 and 1.0C, the LFP/PZL/Li cell has a smaller polarization voltage than the LFP/PL/Li cells. What is more, LFP/PZL/Li also has better

cycling performance (Figure 6c). The initial discharge capacity was $157.8 \text{ mA h g}^{-1}$ at 0.2C, and the capacity retention rate was 93.03% after 200 cycles. Also, Figure S7 shows the charge/discharge curves of LFP/PL/Li versus LFP/PZL/Li cells before and after cycling for 110 cycles. The improved electrochemical performance of the LFP/PZL/Li cell is attributed to the higher IC and Li^+ transference numbers of the PZL electrolyte, which can promote ion transport and reduce the polarization of the cell.⁴⁶ Furthermore, the distribution of ZIF-8 can effectively immobilize anions, which is favorable for Li^+ transport, while the excellent mechanical properties enable it to prevent the growth of Li dendrites, thus ensuring the uniform deposition of Li and the stability of the interface, which is also manifested in the excellent cycling stability of the LFP/PZL/Li cells.^{26,49} As shown in Figure S8, in the FESEM images of the disassembled

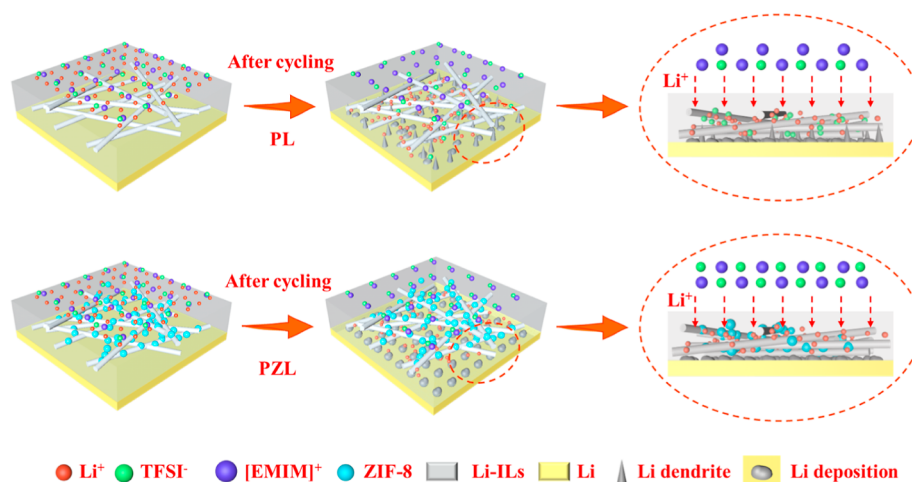


Figure 5. Schematic diagram of Li/Li symmetrical cells before and after cycling.

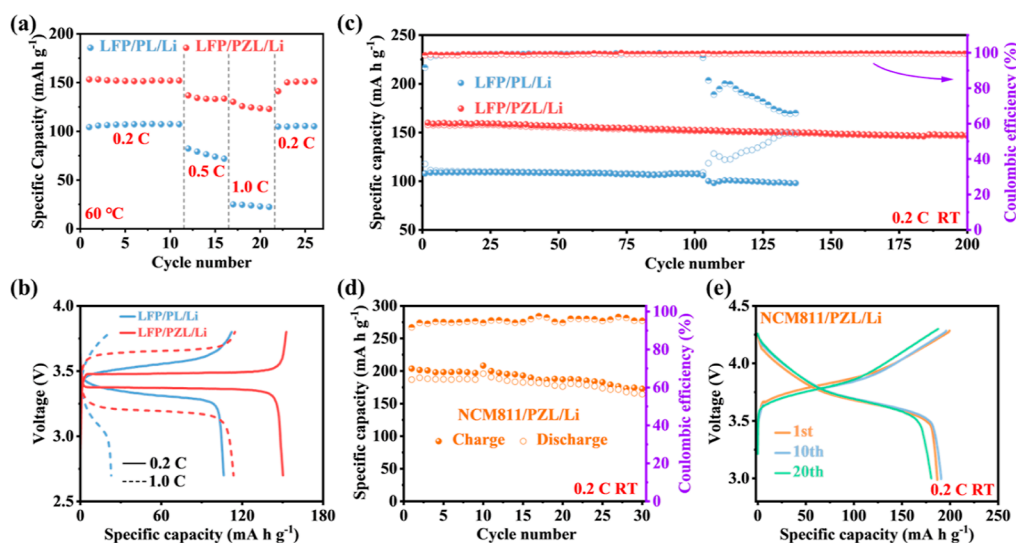


Figure 6. Performance of LFP/PL/Li and LFP/PZL/Li cells. (a) Rate performance. (b) Charge/discharge curves at 0.2 and 1C. (c) Cycling performance at 0.2C. (d) Cycling performance and Coulombic efficiency at 0.2C. (e) Charge and discharge voltage profiles of the NCM811/PZL/Li cells.

cycled LFP/PZL/Li cell test PZL, the cycling PZL electrolyte can still be seen with a clear fiber network and no obvious cracks, which indicates that the PZL electrolyte has a good structural stability. Moreover, $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}(\text{NCM811})/\text{PZL}/\text{Li}$ cells containing ternary electrodes were assembled and examined for their cycling performance, as shown in Figure 6d,e. The NCM811/PZL/Li cells had an initial discharge capacity of $186.7 \text{ mA h g}^{-1}$ and a specific capacity of $168.6 \text{ mA h g}^{-1}$ after cycling for 30 revolutions at 0.2C. This indicates that the PZL electrolyte has the potential to be compatible with high-voltage cathodes.

3. CONCLUSIONS

In summary, a SLE based on Li-ILs-impregnated PVDF-HFP/ZIF-8 composite membranes was proposed in this paper. The electrolyte has a special structure of PVDF-HFP fibers loaded with ZIF-8 particles. Benefiting from this structure and the impregnation of Li-ILs, the IC ($2.29 \times 10^{-3} \text{ S cm}^{-1}$, 30 °C), Li^+ transference number (0.56), and mechanical properties of the PZL SLEs were improved. The results show that due to the stable interfacial situation between PZL and Li metal, the Li/

PZL/Li cells were able to exhibit highly stable plating/stripping performance at 0.2 mA cm^{-2} for 2000 h. The $\text{LiFePO}_4/\text{Li}$ cells assembled using PZL exhibit excellent cycling performance, as evidenced by an initial discharge capacity of $157.8 \text{ mA h g}^{-1}$ and a capacity retention of 93.03% after 200 cycles at 0.2C. This work provides an excellent strategy for the design of MOF-based solid electrolytes for high-safety LMBs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.energyfuels.3c02989>.

Additional experimental details, materials, and methods; FESEM images of ZIF-8 with BET data; Nyquist plots and EIS curves; Li^+ transference number of PL; and FESEM images of PZL after cycling (PDF)

AUTHOR INFORMATION

Corresponding Authors

Ying Huang – MOE Key Laboratory of Material Physics and Chemistry under Extraordinary Conditions, School of Chemistry and Chemical Engineering Northwestern Polytechnical University, Xi'an 710072, PR China; orcid.org/0000-0001-5677-8426; Email: yingh@nwpu.edu.cn

Zheng Zhang – Beijing Key Laboratory of Membrane Materials and Engineering, Department of Chemical Engineering, Tsinghua University, 100084 Beijing, China; Email: zhengzhang0527@163.com

Authors

Wanqing Fan – MOE Key Laboratory of Material Physics and Chemistry under Extraordinary Conditions, School of Chemistry and Chemical Engineering Northwestern Polytechnical University, Xi'an 710072, PR China

Meirong Gu – Shanghai Institute of Space Power-sources, Shanghai 200245, PR China

Xu Lu – Shanghai Institute of Space Power-sources, Shanghai 200245, PR China

Kaihong She – MOE Key Laboratory of Material Physics and Chemistry under Extraordinary Conditions, School of Chemistry and Chemical Engineering Northwestern Polytechnical University, Xi'an 710072, PR China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.energyfuels.3c02989>

Notes

The authors declare no competing financial interest.

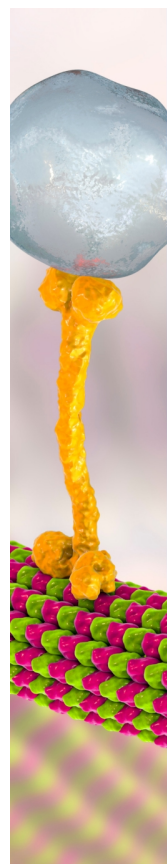
ACKNOWLEDGMENTS

The authors thank the Analysis and Testing Center of Northwestern Polytechnical University for their technical assistance. The authors gratefully acknowledge financial support from the China Postdoctoral Science Foundation (2022TQ0173 and 2023M731922).

REFERENCES

- (1) Liu, S.; Ji, X.; Piao, N.; Chen, J.; Eidson, N.; Xu, J.; Wang, P.; Chen, L.; Zhang, J.; Deng, T.; et al. An Inorganic-Rich Solid Electrolyte Interphase for Advanced Lithium-Metal Batteries in Carbonate Electrolytes. *Angew. Chem., Int. Ed. Engl.* **2021**, *60* (7), 3661–3671.
- (2) Wu, J.; Shen, L.; Zhang, Z.; Liu, G.; Wang, Z.; Zhou, D.; Wan, H.; Xu, X.; Yao, X. All-Solid-State Lithium Batteries with Sulfide Electrolytes and Oxide Cathodes. *Electrochem. Energy Rev.* **2021**, *4* (1), 101–135.
- (3) Zhang, Z.; Ying, H.; Zhang, G.; Zhang, G.; Chao, L. Three-dimensional fiber network reinforced polymer electrolyte for dendrite-free all-solid-state lithium metal batteries. *Energy Storage Mater.* **2021**, *41*, 631–641.
- (4) Zhang, Z.; Huang, Y.; Li, C.; Li, X. Metal-Organic Framework-Supported Poly(ethylene oxide) Composite Gel Polymer Electrolytes for High-Performance Lithium/Sodium Metal Batteries. *ACS Appl. Mater. Interfaces* **2021**, *13* (31), 37262–37272.
- (5) Dirican, M.; Yan, C.; Zhu, P.; Zhang, X. Composite solid electrolytes for all-solid-state lithium batteries. *Mater. Sci. Eng., R* **2019**, *136*, 27–46.
- (6) Wu, J.; Liu, S.; Han, F.; Yao, X.; Wang, C. Lithium/Sulfide All-Solid-State Batteries using Sulfide Electrolytes. *Adv. Mater.* **2021**, *33* (6), No. e2000751.
- (7) Sun, C.; Zhang, J. H.; Yuan, X. F.; Duan, J. N.; Deng, S. W.; Fan, J. M.; Chang, J. K.; Zheng, M. S.; Dong, Q. F. ZIF-8-Based Quasi-Solid-State Electrolyte for Lithium Batteries. *ACS Appl. Mater. Interfaces* **2019**, *11* (50), 46671–46677.
- (8) Han, S. A.; Qutaish, H.; Lee, J. W.; Park, M. S.; Kim, J. H. Metal-organic framework derived porous structures towards lithium rechargeable batteries. *EcoMat* **2023**, *5* (2), No. e12283.
- (9) Ye, Z.; Jiang, Y.; Li, L.; Wu, F.; Chen, R. Rational Design of MOF-Based Materials for Next-Generation Rechargeable Batteries. *Nano-Micro Lett.* **2021**, *13* (1), 203.
- (10) Xia, Y.; Xu, N.; Du, L.; Cheng, Y.; Lei, S.; Li, S.; Liao, X.; Shi, W.; Xu, L.; Mai, L. Rational Design of Ion Transport Paths at the Interface of Metal-Organic Framework Modified Solid Electrolyte. *ACS Appl. Mater. Interfaces* **2020**, *12* (20), 22930–22938.
- (11) Dong, R.; Zheng, J.; Yuan, J.; Li, Y.; Zhang, T.; Liu, Y.; Liu, Y.; Sun, Y.; Zhong, B.; Chen, Y.; et al. A polyethylene oxide/metal-organic framework composite solid electrolyte with uniform Li deposition and stability for lithium anode by immobilizing anions. *J. Colloid Interface Sci.* **2022**, *620*, 47–56.
- (12) Zheng, Y.; Yang, N.; Gao, R.; Li, Z.; Dou, H.; Li, G.; Qian, L.; Deng, Y.; Liang, J.; Yang, L.; et al. "Tree-Trunk" Design for Flexible Quasi-Solid-State Electrolytes with Hierarchical Ion-Channels Enabling Ultralong-Life Lithium-Metal Batteries. *Adv. Mater.* **2022**, *34* (44), No. e2203417.
- (13) Huo, H.; Wu, B.; Zhang, T.; Zheng, X.; Ge, L.; Xu, T.; Guo, X.; Sun, X. Anion-immobilized polymer electrolyte achieved by cationic metal-organic framework filler for dendrite-free solid-state batteries. *Energy Storage Mater.* **2019**, *18*, 59–67.
- (14) Chen, L.; Luque, R.; Li, Y. Controllable design of tunable nanostructures inside metal-organic frameworks. *Chem. Soc. Rev.* **2017**, *46* (15), 4614–4630.
- (15) Wang, Z.; Wang, D.; Zhang, S.; Hu, L.; Jin, J. Interfacial Design of Mixed Matrix Membranes for Improved Gas Separation Performance. *Adv. Mater.* **2016**, *28* (17), 3399–3405.
- (16) Zhu, H.; Yang, X.; Cranston, E. D.; Zhu, S. Flexible and Porous Nanocellulose Aerogels with High Loadings of Metal-Organic-Framework Particles for Separations Applications. *Adv. Mater.* **2016**, *28* (35), 7652–7657.
- (17) Zhao, X.; Zhu, M.; Tang, C.; Quan, K.; Tong, Q.; Cao, H.; Jiang, J.; Yang, H.; Zhang, J. ZIF-8@MXene-reinforced flame-retardant and highly conductive polymer composite electrolyte for dendrite-free lithium metal batteries. *J. Colloid Interface Sci.* **2022**, *620*, 478–485.
- (18) Lei, Z.; Shen, J.; Zhang, W.; Wang, Q.; Wang, J.; Deng, Y.; Wang, C. Exploring porous zeolitic imidazolate frame work-8 (ZIF-8) as an efficient filler for high-performance poly(ethyleneoxide)-based solid polymer electrolytes. *Nano Res.* **2020**, *13* (8), 2259–2267.
- (19) Jiang, S.; Lv, T.; Peng, Y.; Pang, H. MOFs Containing Solid-State Electrolytes for Batteries. *Adv. Sci.* **2023**, *10* (10), No. e2206887.
- (20) Han, Q.; Wang, S.; Jiang, Z.; Hu, X.; Wang, H. Composite Polymer Electrolyte Incorporating Metal-Organic Framework Nano-sheets with Improved Electrochemical Stability for All-Solid-State Li Metal Batteries. *ACS Appl. Mater. Interfaces* **2020**, *12* (18), 20514–20521.
- (21) Shen, L.; Wu, H. B.; Liu, F.; Brosmer, J. L.; Shen, G.; Wang, X.; Zink, J. I.; Xiao, Q.; Cai, M.; Wang, G.; et al. Creating Lithium-Ion Electrolytes with Biomimetic Ionic Channels in Metal-Organic Frameworks. *Adv. Mater.* **2018**, *30* (23), No. e1707476.
- (22) Wiers, B. M.; Foo, M. L.; Balsara, N. P.; Long, J. R. A solid lithium electrolyte via addition of lithium isopropoxide to a metal-organic framework with open metal sites. *J. Am. Chem. Soc.* **2011**, *133* (37), 14522–14525.
- (23) Qi, X.; Cai, D.; Wang, X.; Xia, X.; Gu, C.; Tu, J. Ionic Liquid-Imregnated ZIF-8/Polypropylene Solid-like Electrolyte for Dendrite-free Lithium-Metal Batteries. *ACS Appl. Mater. Interfaces* **2022**, *14* (5), 6859–6868.
- (24) Wang, Z.; Tan, R.; Wang, H.; Yang, L.; Hu, J.; Chen, H.; Pan, F. A Metal-Organic-Framework-Based Electrolyte with Nanowetted Interfaces for High-Energy-Density Solid-State Lithium Battery. *Adv. Mater.* **2018**, *30* (2), 1704436.

- (25) Pan, J.; Wang, N.; Fan, H. J. Gel Polymer Electrolytes Design for Na-Ion Batteries. *Small Methods* **2022**, *6* (11), No. e2201032.
- (26) Hao, Z.; Wu, Y.; Zhao, Q.; Tang, J.; Zhang, Q.; Ke, X.; Liu, J.; Jin, Y.; Wang, H. Functional Separators Regulating Ion Transport Enabled by Metal-Organic Frameworks for Dendrite-Free Lithium Metal Anodes. *Adv. Funct. Mater.* **2021**, *31* (33), 2102938.
- (27) Wang, G.; He, P.; Fan, L. Z. Asymmetric Polymer Electrolyte Constructed by Metal-Organic Framework for Solid-State, Dendrite-Free Lithium Metal Battery. *Adv. Funct. Mater.* **2021**, *31* (3), 2007198.
- (28) Watanabe, M.; Thomas, M. L.; Zhang, S.; Ueno, K.; Yasuda, T.; Dokko, K. Application of Ionic Liquids to Energy Storage and Conversion Materials and Devices. *Chem. Rev.* **2017**, *117* (10), 7190–7239.
- (29) Fujie, K.; Ikeda, R.; Otsubo, K.; Yamada, T.; Kitagawa, H. Lithium Ion Diffusion in a Metal-Organic Framework Mediated by an Ionic Liquid. *Chem. Mater.* **2015**, *27* (21), 7355–7361.
- (30) Qiu, B.; Liang, K.; Huang, W.; Zhang, G.; He, C.; Zhang, P.; Mi, H. Crystal-Facet Manipulation and Interface Regulation via TMP-Modulated Solid Polymer Electrolytes toward High-Performance Zn Metal Batteries. *Adv. Energy Mater.* **2023**, *13*, 32–2301193.
- (31) Qiu, B.; Xu, F.; Qiu, J.; Yang, M.; Zhang, G.; He, C.; Zhang, P.; Mi, H.; Ma, J. Electrode-electrolyte interface mediation via molecular anchoring for 4.7 V quasi-solid-state lithium metal batteries. *Energy Storage Mater.* **2023**, *60*, 102832.
- (32) Wu, W.; Dong, S.; Zhang, X.; Hao, J. Gel electrolytes and aerogel electrodes from ILs-based emulsions for supercapacitor applications. *Chem. Eng. J.* **2022**, *446*, 137328.
- (33) Liu, W.; Mi, Y.; Weng, Z.; Zhong, Y.; Wu, Z.; Wang, H. Functional metal-organic framework boosting lithium metal anode performance via chemical interactions. *Chem. Sci.* **2017**, *8* (6), 4285–4291.
- (34) Moh, P. Y.; Cubillas, P.; Anderson, M. W.; Attfield, M. P. Revelation of the molecular assembly of the nanoporous metal organic framework ZIF-8. *J. Am. Chem. Soc.* **2011**, *133* (34), 13304–13307.
- (35) Chen, J.; Huang, L.; Wang, Q.; Wu, W.; Zhang, H.; Fang, Y.; Dong, S. Bio-inspired nanozyme: a hydratase mimic in a zeolitic imidazolate framework. *Nanoscale* **2019**, *11* (13), 5960–5966.
- (36) Xu, F.; Deng, S.; Guo, Q.; Zhou, D.; Yao, X. Quasi-Ionic Liquid Enabling Single-Phase Poly(vinylidene fluoride)-Based Polymer Electrolytes for Solid-State LiNi(0.6) Co(0.2) Mn(0.2) O(2) ||Li Batteries with Rigid-Flexible Coupling Interphase. *Small Methods* **2021**, *5* (7), No. e2100262.
- (37) Choi, H.; Kim, H. W.; Ki, J.-K.; Lim, Y. J.; Kim, Y.; Ahn, J.-H. Nanocomposite quasi-solid-state electrolyte for high-safety lithium batteries. *Nano Res.* **2017**, *10* (9), 3092–3102.
- (38) Zhang, H.; Cao, G.; Yang, Y.; Gu, Z. Capacitive performance of an ultralong aligned carbon nanotube electrode in an ionic liquid at 60°C. *Carbon* **2008**, *46* (1), 30–34.
- (39) Li, Z.; Wang, S.; Shi, J.; Liu, Y.; Zheng, S.; Zou, H.; Chen, Y.; Kuang, W.; Ding, K.; Chen, L.; et al. A 3D interconnected metal-organic framework-derived solid-state electrolyte for dendrite-free lithium metal battery. *Energy Storage Mater.* **2022**, *47*, 262–270.
- (40) Guo, J.; Feng, F.; Zhao, S.; Wang, R.; Yang, M.; Shi, Z.; Ren, Y.; Ma, Z.; Chen, S.; Liu, T. Achieving Ultra-Stable All-Solid-State Sodium Metal Batteries with Anion-Trapping 3D Fiber Network Enhanced Polymer Electrolyte. *Small* **2023**, *19* (16), No. e2206740.
- (41) Liu, X.; Liang, Q.; Chen, L.; Tang, J.; Liu, J.; Tang, M.; Wang, Z. PEO-Based Solid-State Electrolytes Reinforced by High Strength, Interconnected MOF Networks. *ACS Appl. Energy Mater.* **2023**, *6* (9), 4881–4891.
- (42) He, W.; Ding, H.; Chen, X.; Yang, W. Three-dimensional LLZO/PVDF-HFP fiber network-enhanced ultrathin composite solid electrolyte membrane for dendrite-free solid-state lithium metal batteries. *J. Membr. Sci.* **2023**, *665*, 121095.
- (43) Sheng, O.; Zheng, J.; Ju, Z.; Jin, C.; Wang, Y.; Chen, M.; Nai, J.; Liu, T.; Zhang, W.; Liu, Y.; et al. In Situ Construction of a LiF-Enriched Interface for Stable All-Solid-State Batteries and its Origin Revealed by Cryo-TEM. *Adv. Mater.* **2020**, *32* (34), No. e2000223.
- (44) Deng, T.; Cao, L.; He, X.; Li, A.-M.; Li, D.; Xu, J.; Liu, S.; Bai, P.; Jin, T.; Ma, L.; et al. In situ formation of polymer-inorganic solid-electrolyte interphase for stable polymeric solid-state lithium-metal batteries. *Chem* **2021**, *7* (11), 3052–3068.
- (45) Su, Y.; Zhang, X.; Du, C.; Luo, Y.; Chen, J.; Yan, J.; Zhu, D.; Geng, L.; Liu, S.; Zhao, J.; et al. An All-Solid-State Battery Based on Sulfide and PEO Composite Electrolyte. *Small* **2022**, *18* (29), No. e2202069.
- (46) Cho, S. K.; Oh, K. S.; Shin, J. C.; Lee, J. E.; Lee, K. M.; Cho, J.; Lee, W. B.; Kwak, S. K.; Lee, M.; Lee, S. Y. Anion-Rectifying Polymeric Single Lithium-Ion Conductors. *Adv. Funct. Mater.* **2022**, *32* (6), 2107753.
- (47) Li, C.; Deng, S.; Feng, W.; Cao, Y.; Bai, J.; Tian, X.; Dong, Y.; Xia, F. A Universal Room-Temperature 3D Printing Approach Towards porous MOF Based Dendrites Inhibition Hybrid Solid-State Electrolytes. *Small* **2023**, *19* (21), No. e2300066.
- (48) Yang, Q.; Li, G.; Shi, D.; Gao, L.; Deng, N.; Kang, W.; Cheng, B. Composite Solid Electrolyte with Continuous and Fast Organic-Inorganic Ion Transport Highways Created by 3D Crimped Nanofibers@functional Ceramic Nanowires. *Small* **2023**, *19*, No. e2301521.
- (49) Cui, S.; Wu, X.; Yang, Y.; Fei, M.; Liu, S.; Li, G.; Gao, X.-P. Heterostructured Gel Polymer Electrolyte Enabling Long-Cycle Quasi-Solid-State Lithium Metal Batteries. *ACS Energy Lett.* **2022**, *7* (1), 42–52.



CAS BIOFINDER DISCOVERY PLATFORM™

**PRECISION DATA
FOR FASTER
DRUG
DISCOVERY**

CAS BioFinder helps you identify
targets, biomarkers, and pathways

Unlock insights

CAS
A Division of the
American Chemical Society